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SAFETY MECHANISM FOR ACCESSIBLE LIFT EQUIPPED VEHICLES

Abstract

Disclosed is a vehicle rack comprising a bed mount configured to be mounted on a vehicle bed, the bed mount comprising: a mounting plate including a first portion adjacent to an inner wall of the vehicle bed, a second portion adjacent to an outer wall of the vehicle bed, and a third portion connecting the first and second portions, at least two mounting shaft sleeves attached to the second portion of the mounting plate, and a receptacle configured to be detachably connected to the bed mount, the receptacle comprising: at least two mounting shafts, each configured to be inserted into a corresponding one of the at least two mounting shaft sleeves, a plurality of pairs of through holes in the mounting shafts, and at least one stop pin configured to be positioned within at least one pair of the plurality of pairs of through holes.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of priority under 35 U.S.C. § 119 to U.S. Provisional Application No. 63/451,492, filed Mar. 10, 2023, in the United States Patent and Trademark Office, the entire contents of which is hereby incorporated by reference.

FIELD OF THE INVENTION

[0002] This disclosure relates to an apparatus for accessible lift-equipped vehicles and, more particularly, to a safety mechanism for accessible lift-equipped vehicles.

BACKGROUND

[0003] Vehicular wheelchair lifts are widely utilized to assist persons in wheelchairs to get into and out of vehicles (e.g., vans, shuttles, buses, etc.). These lifts typically have a moveable platform for transporting an occupied wheelchair, where the moveable platform can be raised and lowered between a position at a ground level outside a vehicle and a position at a floor level inside the vehicle. However, when moveable platform is at a ground level, a wheelchair at floor level within the vehicle may roll or be rolled out of the vehicle.

[0004] The disclosed embodiments have been provided to address these and other shortcomings.

SUMMARY

[0005] In some exemplary embodiments, the disclosure is directed to a lift safety apparatus, comprising: a fixed plate configured to be removably attached within a vehicle; a lift plate adjacent to the fixed plate; and a hinge connecting the lift plate to the fixed plate, wherein the lift plate comprises first and second arms, and wherein each of the first and second arms comprises a first vertical arm portion, a second vertical arm portion, and a contact arm portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The foregoing summary, as well as the following detailed description of example embodiments of this invention, will be better understood when read in conjunction with the appended drawings. In the drawings:

[0007] FIG. 1 is a plan view of a lift safety mechanism prior to fabrication, according to certain example embodiments;

[0008] FIG. 2 is a side view of an arm of the lift safety mechanism of FIG. 1, according to certain example embodiments;

[0009] FIG. 3 is a front view of an arm of the lift safety mechanism of FIG. 1, according to certain example embodiments;

[0010] FIG. 4 is an enlarged plan view of an arm of the lift safety mechanism of FIG. 1, according to certain example embodiments; and

[0011] FIG. 5 is a view illustrating attachment and operation of lift safety mechanism, according to certain example embodiments.

DETAILED DESCRIPTION

[0012] The present disclosure now will be described more fully hereinafter with reference to the accompanying drawings, in which various embodiments are shown. The invention may, however, be embodied in many different forms and should not be construed as limited to the example embodiments set forth herein. These example embodiments are just that—examples—and many implementations and variations are possible that do not require the details provided herein. It should also be emphasized that the disclosure provides details of alternative examples, but such listing of alternatives is not exhaustive. Furthermore, any consistency of detail between various examples should not be interpreted as requiring such detail—it is impracticable to list every possible variation for every feature described herein. The language of the claims should be

referenced in determining the requirements of the invention.

[0013] In the drawings, the size and relative sizes of features and components may be exaggerated for clarity. Like reference characters refer to like elements throughout. Though the different figures show variations of exemplary embodiments, these figures are not necessarily intended to be mutually exclusive from each other. Rather, as will be seen from the context of the detailed description below, certain features depicted and described in different figures can be combined with other features from other figures to result in various embodiments, when taking the figures and their description as a whole into consideration. The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention.

[0014] It will be understood that when an element is referred to as being “connected” or “coupled” to or “on” another element, it can be directly connected or coupled to or on the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, or as “contacting” or “in contact with” another element (or using any form of the word “contact”), there are no intervening elements present at the point of contact.

[0015] Ordinal numbers such as “first,” “second,” “third,” etc. may be used simply as labels of certain elements, steps, etc., to distinguish such elements, steps, etc. from one another. Terms that are not described using “first,” “second,” etc., in the specification, may still be referred to as “first” or “second” in a claim. In addition, a term that is referenced with a particular ordinal number (e.g., “first” in a particular claim) may be described elsewhere with a different ordinal number (e.g., “second” in the specification or another claim).

[0016] FIG. 1 is a plan view of a lift safety mechanism, according to certain example embodiments. FIG. 4 is an enlarged plan view of an arm of the lift safety mechanism of FIG. 1, according to certain example embodiments. FIGS. 1 and 4 are plan views of the lift safety mechanism prior to fabrication (e.g., before the lift plate is bent along the dashed lines in accordance with the bend lines or bend profiles).

[0017] FIG. 2 is a side view of an arm of the lift safety mechanism of FIG. 1, according to certain example embodiments. FIG. 3 is a front view of an arm of the lift safety mechanism of FIG. 1, according to certain example embodiments. FIGS. 2 and 3 illustrate an arm of the lift safety mechanism after fabrication (e.g., after the lift plate is bent along the dashed lines in accordance with the bend lines or bend profiles).

[0018] Referring to FIGS. 1, 2, 3, and 4, the lift safety mechanism **100** may be comprised of at least two plate portions. For example, the lift safety mechanism **100** may include a fixed plate **101** and a lift plate **102**. In example embodiments, each of the fixed plate **101** and the lift plate **102** may be formed of a single piece of rigid material (e.g., a metal) that can be bent, as discussed further below. The rigid material forming each of the fixed plate **101** and the lift plate **102** may be steel, such as, for example, an 11-gauge steel plate. However, those of ordinary skill in the art will appreciate from this disclosure that any suitable material, such as allow, aluminum, composite, or the like, may be used without departing from the scope of the present invention. For example, the aforementioned lift safety mechanism **100** may be constructed of other gauges of steel, carbon fiber, plastic polymer matrices, or other materials.

[0019] The fixed plate **101** may include a first side **101_S1** and a second side **101_S2** opposite the first side **101_S1**. The first side **101_S1** and the second side **101_S2** may extend lengthwise in parallel with one another. The fixed plate **101** may include a first end **101_E1** and a second end **101_E2** opposite the first end **101_E1**. The first end **101_E1** and the second end **101_E2** may connect the first side **101_S1** and the second side **101_S2** of the fixed plate **101**. The first end **101_E1** and the second end **101_E2** may mirror one another. For example, a shape of the first end **101_E1** may mirror a shape of the second end **101_E2**.

[0020] The first end **101_E1** may include a first first end surface **101_E1a**, a second first end surface **101_E1b**, and a connecting first end surface **101_E1c**. The first first end surface **101_E1a**

may be substantially perpendicular to the first side **101_S1**, and the second first end surface **101_E1b** may be substantially perpendicular to the second side **101_S2**. The connecting first end surface **101_E1c** may be directly connected to each of the first first end surface **101_E1a** and the second first end surface **101_E1b**. In example embodiments, the connecting first end surface **101_E1c** may be substantially parallel to each of the first side **101_S1** and the second side **101_S2**, but embodiments are not limited thereto. For example, the connecting first end surface **101_E1c** may be at an angle with respect to the first side **101_S1** and the second side **101_S2**. For example, the connecting first end surface **101_E1c** may not be perpendicular to either of the first first end surface **101_E1a** or the second first end surface **101_E1b**.

[0021] The second end **101_E2** may include a first second end surface **101_E2a**, a second second end surface **101_E2b**, and a connecting second end surface **101_E2c**. The first second end surface **101_E2a** may be substantially perpendicular to the first side **101_S1**, and the second second end surface **101_E2b** may be substantially perpendicular to the second side **101_S2**. The connecting second end surface **101_E2c** may be directly connected to each of the first second end surface **101_E2a** and the second second end surface **101_E2b**. In example embodiments, the connecting second end surface **101_E2c** may be substantially parallel to each of the first side **101_S1** and the second side **101_S2**, but embodiments are not limited thereto. For example, the connecting second end surface **101_E2c** may be at an angle with respect to the first side **101_S1** and the second side **101_S2**. For example, the connecting second end surface **101_E2c** may not be perpendicular to either of the first second end surface **101_E2a** or the second second end surface **101_E2b**.

[0022] The fixed plate **101** may further include a main portion **101_M** between the second side **101_S2** and the bend line **101_BL**, and a beveled edge **101_BE** between the bend line **101_BL** and the first side **101_S1**. The beveled edge **101_BE** may have a first width **W1** between the bend line **101_BL** and the first side **101_S1**. The first width **W1** of the beveled edge **101_BE** may be in the range of about 0.5-1.5 inches. In example embodiments, the first width **W1** of the beveled edge **101_BE** may be about $\frac{3}{4}$ inches.

[0023] The beveled edge **101_BE** may be formed by bending the fixed plate **101** at the bend line **101_BL**. For example, the fixed plate **101** may be bent at the bend line **101_BL** to form the beveled edge **101_BE**. Referring to FIG. 2, the beveled edge **101_BE** may form an angle $\theta 1$ with respect to the main portion **101_M** of the fixed plate **101**. In example embodiments, the angle $\theta 1$ may be in the range of about 30° to about 60° . In some embodiments, the angle $\theta 1$ may be in the range of about 40° to about 50° . In certain embodiments, the angle $\theta 2$ may be about 45° . The beveled edge **101_BE** may function to allow a wheeled device, such as a wheelchair or stretcher, to roll smoothly onto and over the fixed plate **101**.

[0024] The main portion **101_M** may have a first length **L1** between the first first end surface **101_E1a** and the first second end surface **101_E2a**, and a second length **L2** between the first second end surface **101_E1b** and the second second end surface **101_E2b**. The first length **L1** may be less than the second length **L2**. The second length **L2** may be in the range of about 1-3 inches larger than the first length **L1**. The first length **L1** may be in a range of about 34-39 inches, and the second length **L2** may be in a range of about 35-42 inches. In some embodiments, the first length **L1** may be in a range of about 35-36 inches, and the second length **L2** may be in a range of about 37-38 inches. For example, the first length **L1** be about $35\frac{1}{8}$ inches, and the second length **L2** may be about $37\frac{1}{4}$ inches. As another example, the first length **L1** may be about $38\frac{1}{8}$ inches, and the second length **L2** may be about $40\frac{1}{4}$ inches.

[0025] The main portion **101_M** of the fixed plate **101** may have a second width **W2**, which is the width of the fixed plate **101** from the bend line **101_BL** to the second side **101_S2** of the fixed plate **101**. The main portion **101_M** may include a first main portion **101_M1** and a second main portion **101_M2**. The first main portion **101_M1** may be adjacent to the beveled edge **101_BE** and may have the third width **W3**, which is the width between the bend line **101_BL** and the transition from the first length **L1** to the second length **L2**. The second main portion **101_M2** may be adjacent to

the second side **101_S2** of the fixed plate **101**. The second main portion **101_M2** may have a fourth width **W4**, which is the width between second side **101_S2** of the fixed plate **101** and the transition from the first length **L1** to the second length **L2**. The second width **W2** may be equal to the sum of the third width **W3** and the fourth width **W4**. The second main portion **101_M2** may have the second length **L2**, and the first main portion **101_M1** may have the first length **L1**, which is smaller than the second length **L2**. In some embodiments, the connecting first end surface **101_E1c** and the connecting second end surface **101_E2c** may define the transition from the first length **L1** to the second length **L2**.

[0026] As discussed further below in connection with FIG. 5, the fixed plate **101** may straddle or conform to components of a lift mechanism installed in a transportation vehicle. In example embodiments, the fixed plate may be disposed between arms of a lift mechanism previously installed on the transportation vehicle and/or on a lift pressure plate provided on a floor of the transportation vehicle. For example, the fixed plate **101** may be provided on top of or over an existing pressure plate of the transportation vehicle. The fixed plate **101** may lay substantially parallel to the floor of the transportation vehicle. The fixed plate **101** may be affixed to the transportation vehicle using pressure plate fasteners extending inserted through holes **101_H** in the fixed plate **101**.

[0027] FIG. 1 illustrates a single hole **101_H** adjacent to a first end of the fixed plate **101**, and a single hole **101_H** adjacent to a second end **101_E2** of the fixed plate **101**. However, embodiments are not limited thereto, and there may be a plurality of holes **101_H** adjacent to each of the first end **101_E1** and a second end **101_E2** of the fixed plate **101**. The holes **101_H** may be in the range of $\frac{1}{4}$ to $\frac{3}{4}$ inches in diameter. In some embodiments, each hole **101_H** may be about $\frac{3}{8}$ inches in diameter. The pressure plate fasteners may be, for example, bolts, rods, screws, etc.

[0028] In some embodiments, the intersection of the first side **101_S1** and the first end **101_E1** may be rounded, and the intersection of the first side **101_S1** and the second end **101_E2** may be rounded. For example, the intersection of the first side **101_S1** and the first end **101_E1** and the intersection of the first side **101_S1** and the second end **101_E2** each may be rounded corners having a radius of about 1 inch. However, embodiments are not limited thereto. In some embodiments, the first side **101_S1** and the first end **101_E1** may form a right angle with one another, and the first side **101_S1** and the second end **101_E2** may form a right angle with one another.

[0029] In some embodiments, the intersection of the second side **101_S2** and the first end **101_E1** may be rounded, and the intersection of the second side **101_S2** and the second end **101_E2** may be rounded. For example, the intersection of the second side **101_S2** and the first end **101_E1** and the intersection of the second side **101_S2** and the second end **101_E2** each may be a rounded corner having a radius of about 1 inch. However, embodiments are not limited thereto. In some embodiments, the second side **101_S2** and the first end **101_E1** may form a right angle with one another, and the second side **101_S2** and the second end **101_E2** may form a right angle with one another.

[0030] In some embodiments, the intersection of the connecting first end surface **101_E1c** and the first first end surface **101_E1a** may be rounded, and the intersection of the connecting first end surface **101_E1c** and the second first end surface **101_E1b** may be rounded. For example, the intersection of the connecting first end surface **101_E1c** and the first first end surface **101_E1a** and the intersection of the connecting first end surface **101_E1c** and the second first end surface **101_E1b** each may be a rounded corner having a radius of about 1 inch. However, embodiments are not limited thereto. In some embodiments, the intersection of the connecting first end surface **101_E1c** and the first first end surface **101_E1a** may form a right angle, and the intersection of the connecting first end surface **101_E1c** and the second first end surface **101_E1b** may form a right angle.

[0031] In some embodiments, the intersection of the connecting second end surface **101_E2c** and

the first second end surface **101_E2a** may be rounded, and the intersection of the connecting second end surface **101_E2c** and the second second end surface **101_E2b** may be rounded. For example, the intersection of the connecting second end surface **101_E2c** and the first second end surface **101_E2a** and the intersection of the connecting second end surface **101_E2c** and the second second end surface **101_E2b** each may be a rounded corner having a radius of about 1 inch. However, embodiments are not limited thereto. In some embodiments, the intersection of the connecting second end surface **101_E2c** and the first second end surface **101_E2a** may form a right angle, and the intersection of the connecting second end surface **101_E2c** and the second second end surface **101_E2b** may form a right angle.

[0032] The lift plate **102** may include a first side **102_S1** and a second side **102_S2** opposite the first side **102_S1**. The first side **102_S1** and the second side **102_S2** may extend lengthwise in parallel with one another. The lift plate **102** may include a first end **102_E1** and a second end **102_E2** opposite the first end **102_E1**. The lift plate **102** may have a fifth width **W5** between the first side **102_S1** and the second side **102_S2**. The fifth width **W5** may be in a range of about 5 to 9 inches. In example embodiments, the fifth width **W5** may be in a range of about 6 to 8 inches. In some embodiments, the fifth width **W5** may be about 7 inches.

[0033] The lift plate **102** may further include a main portion **102_M** between the second side **102_S2** and the bend line **102_BL**, and a beveled edge **102_BE** between the bend line **102_BL** and the first side **102_S1**. The beveled edge **102_BE** may have a sixth width **W6** between the bend line **102_BL** and the first side **102_S1**. The sixth width **W6** of the beveled edge **102_BE** may be in the range of about 0.5-1.5 inches. In example embodiments, the sixth width **W6** of the beveled edge **102_BE** may be about $\frac{5}{8}$ inches.

[0034] The beveled edge **102_BE** may be formed by bending the lift plate **102** at the bend line **102_BL**. For example, the lift plate **102** may be bent at the bend line **102_BL** to form the beveled edge **102_BE**. Referring to FIG. 2, the beveled edge **102_BE** may form an angle $\theta 2$ with respect to the main portion **102_M** of the lift plate **102**. In example embodiments, the angle $\theta 2$ may be in the range of about 5° to about 30° . In some embodiments, the angle $\theta 2$ may be in the range of about 8° to about 12° . In certain embodiments, the angle $\theta 2$ may be about 10° .

[0035] The main portion **102_M** may have a seventh width **W7** between the first side **102_S1** and the bend line **102_BL2**. The seventh width **W7** may be in a range of about 5 to 8 inches. In example embodiments, the seventh width **W7** may be in a range of about 6 to 7 inches. In some embodiments, the seventh width **W7** may be about $6\frac{3}{8}$ inches. The fifth width **W5** may be equal to the sum of the sixth width **W6** and the seventh width **W7**. The main portion **102_M** and the beveled edge **102_BE** each may have the first length **L1** between the first end **102_E1** and the second end **102_E2**.

[0036] The lift plate **102** may be connected to the fixed plate **101** by a hinge **105**. In example embodiments, the hinge **105** may be a single hinge that extends more than about 50% of the first length **L1**. For example, the single hinge **105** may extend more than about 75% of the first length **L1**. More specifically, the single hinge **105** may extend more than about 90% of the first length **L1**. In other example embodiments, the hinge **105** may be a plurality of hinges that extend more than about 50% of the first length **L1**. For example, the plurality of hinges **105** may extend more than about 75% of the first length **L1**. More specifically, the plurality of hinges **105** may extend more than about 80% of the first length **L1**.

[0037] The hinge **105** may have a first part **105A** that overlaps the fixed plate **101** and a second part **105B** that overlaps the lift plate **102**. The first part **105A** may have an eighth width **W8**, and the second part **105B** may have a ninth width **W9**. Each of eighth width **W8** and the ninth width **W9** may be in a range of about $\frac{1}{2}$ to 2 inches. In example embodiments, each of eighth width **W8** and the ninth width **W9** may be about 1 inch.

[0038] The first part **105A** may be attached to an underside of the fixed plate **101**, and the second part **105B** may be attached to an underside of the lift plate **102**. For example, the first part **105A**

may be attached to the underside of the fixed plate **101** by rivets extending through the first part **105A** and the fixed plate **101**, and the second part **105B** may be attached to an underside of the lift plate **102** by rivets extending through the first part **105A** and the lift plate **102**. Alternatively and/or additionally, the first part **105A** may be attached to the underside of the fixed plate **101** by welding the first part **105A** to the fixed plate **101**, and the second part **105B** may be attached to an underside of the lift plate **102** by welding the first part **105A** to the lift plate **102**. However, embodiments are not limited thereto, and the first part **105A** and the second part **105B** may be attached to the fixed plate **101** and the second part **105B** by any suitable mechanism.

[0039] The lift plate **102** may further include a first lift arm **103A** at the first end **102_E1** of the lift plate **102**, and a second lift arm **103B** at the second end **102_E2** of the lift plate **102**. For example, the lift plate **102** may be bent at the first arm bend line **102_BL1a** to form the first arm **103A**, and the lift plate **102** may be bent at the second arm bend line **102_BL2a** to form the second arm **103B**. The first arm **103A** may form a right angle with respect to the main portion **102_M** of the lift plate **102** at the first arm bend line **102_BL1a**, and the second arm **103B** may form a right angle with respect to the main portion **102_M** of the lift plate **102** at the second arm bend line **102_BL2a**. Each of the first arm bend line **102_BL1a** and the second arm bend line **102_BL2a** may be in the range of about 3 to 5 inches. In example embodiments, each of the first arm bend line **102_BL1a** and the second arm bend line **102_BL2a** may be about 4 inches. The first arm **103A** and the second arm **103B** may mirror one another.

[0040] The first arm **103A** may include a first vertical arm portion **103A_v**, a first connecting arm portion **103A_h**, and a first contact arm portion **103A_c**. The first connecting arm portion **103A_h** may form a right angle with respect to the first vertical arm portion **103A_v** at a third bend line **102_BL1b**. The first contact arm portion **103A_c** may form an angle θ_3 with respect to the first connecting arm portion **103A_h** at a fourth bend line **102_BL1c**. The angle θ_3 may be in a range of about 40° to about 60° . In some embodiments, the angle θ_3 may be in the range of about 45° to about 55° . In certain embodiments, the angle θ_3 may be about 50° . The third bend line **102_BL1b** may be in a range of about 3 to 6 inches. In example embodiments, the third bend line **102_BL1b** may be in a range of about 4 to 5 inches. In some embodiments, the third bend line **102_BL1b** may be about $4\frac{1}{8}$ inches. The fourth bend line **102_BL1c** may be in a range of about 1 to 3 inches. In example embodiments, the fourth bend line **102_BL1c** may be in a range of about 1 to 2 inches. In some embodiments, the fourth bend line **102_BL1c** may be about $1\frac{1}{2}$ inches.

[0041] The second arm **103B** may include a second vertical arm portion **103B_v**, a second connecting arm portion **103B_h**, and a second contact arm portion **103B_c**. The second connecting arm portion **103B_h** may form a right angle with respect to the second vertical arm portion **103B_v** at a fifth bend line **102_BL2b**. The second contact arm portion **103B_c** may form an angle θ_3 with respect to the second connecting arm portion **103B_h** at a sixth bend line **102_BL2c**. The angle θ_3 may be in a range of about 40° to about 60° . In some embodiments, the angle θ_3 may be in the range of about 45° to about 55° . In certain embodiments, the angle θ_3 may be about 50° . The fifth bend line **102_BL2b** may be in a range of about 3 to 6 inches. In example embodiments, the fifth bend line **102_BL2b** may be in a range of about 4 to 5 inches. In some embodiments, the fifth bend line **102_BL2b** may be about $4\frac{1}{8}$ inches. The sixth bend line **102_BL2c** may be in a range of about 1 to 3 inches. In example embodiments, the sixth bend line **102_BL2c** may be in a range of about 1 to 2 inches. In some embodiments, the sixth bend line **102_BL2c** may be about $1\frac{1}{2}$ inches.

[0042] Each of the first vertical arm portion **103A_v** and the second vertical arm portion **103B_v** may have a third length **L3** in the range of about 6 to 8 inches. In some embodiments, each of the first arm **103A** and the second arm **103B** may have a third length **L3** of about $7\frac{1}{4}$ inches.

[0043] When bent at the first arm bend line **102_BL1b** and the second arm bend line **102_BL2b**, respectively, each of the first connecting arm portion **103A_h** and the second connecting arm portion **103B_h** may be perpendicular to the upper surface of the lift plate **102**. Bottom surfaces of each of the first connecting arm portion **103A_h** and the second connecting arm portion **103B_h**

may be spaced apart from the lift plate **102** by a fourth length **L4**. The fourth length **L4** may be in the range of about 2 to 5 inches. In some embodiments, the fourth length **L4** may be in the range of about 3 to 4 inches. For example, the fourth length **L4** may be about $3\frac{1}{8}$ inches. Each of the first connecting arm portion **103A_h** and the second connecting arm portion **103B_h** may have a fifth length **L5**. The fifth length **L5** may be in the range of about 3 to 6 inches. In some embodiments, the fifth length **L5** may be in the range of about 4 to 5 inches. For example, the fifth length **L5** may be about $4\frac{1}{8}$ inches. The third length **L3** may be equal to the sum of the fourth length **L4** and the fifth length **L5**.

[0044] Each of the first connecting arm portion **103A_h** and the second connecting arm portion **103B_h** may have an eleventh width **W11** in the range of about 1 to 4 inches. In some embodiments, the eleventh width **W11** may be in the range of about 2 to 3 inches. In example embodiments, the eleventh width **W11** may be about $2\frac{1}{2}$ inches. A bottom surface of the first connecting arm portion **103A_h** may be spaced apart from the lift plate **102** by a fourth length **L4**.

[0045] When bent at the first arm bend line **102_BL1a** and the second arm bend line **102_BL**, respectively, each of the first arm **103A** and the second arm **103B** may have a first height **H1** with respect to the lift plate **102** in a range of about 6 to 8 inches. In some embodiments, each of the first arm **103A** and the second arm **103B** may have a first height **H1** with respect to the lift plate **102** of about $7\frac{1}{4}$ inches. In an example embodiment, each of the first vertical arm portion **103A_v** and the second vertical arm portion **103B_v** may have an upper linear surface, which extends from the lift plate **102** to the first contact arm portion **103A_c** and second contact arm portion **103B_c**, respectively. In addition, each of the first vertical arm portion **103A_v** and the second vertical arm portion **103B_v** may have a lower surface, which extends from the lift plate **102** to the first contact arm portion **103A_c** and the second contact arm portion **103B_c**, respectively. The lower surface may have a linear segment and a curvilinear segment. The first height **H1** may be a sum of a second height **H2**, which is a height from the lift plate **102** to the intersection of the linear segment and the curvilinear segment, and a third height **H3**, which is a height from the intersection of the linear segment and the curvilinear segment to the first contact arm portion **103A_c** and second contact arm portion **103B_c**, respectively. The second height **H2** may be in a range of about 1 to 5 inches. In example embodiments, the second height **H2** may be in a range of about 3 to 4 inches. In some embodiments, the second height **H2** may be about $3\frac{1}{8}$ inches. The third height **H3** may be in a range of about 2 to 6 inches. In example embodiments, the third height **H3** may be in a range of about 3 to 5 inches. In some embodiments, the third height **H3** may be about $4\frac{1}{8}$ inches.

[0046] In example embodiments, each of the first contact arm portion **103A_c** and second contact arm portion **103B_c** may have a sixth length **L6** in the range of about 1 to 4 inches. In some embodiments, each of the first contact arm portion **103A_c** and second contact arm portion **103B_c** may have a sixth length **L6** in the range of about 1 to 3 inches. In example embodiments, each of the first contact arm portion **103A_c** and second contact arm portion **103B_c** may have a sixth length **L6** of about $1\frac{1}{2}$ inches. Each of the first contact arm portion **103A_c** and second contact arm portion **103B_c** may have a tenth width **W10** in the range of about 1 to 4 inches. In some embodiments, each of the first contact arm portion **103A_c** and second contact arm portion **103B_c** may have a tenth width **W10** in the range of about 1 to 3 inches. In example embodiments, each of the first contact arm portion **103A_c** and second contact arm portion **103B_c** may have a tenth width **W10** of about $1\frac{1}{2}$ inches. Although the first contact arm portion **103A_c** and second contact arm portion **103B_c** are illustrated as having a substantially square shape, embodiments are not limited thereto. Each of the first contact arm portion **103A_c** and second contact arm portion **103B_c** may have any polygonal shape (e.g., rectangular, triangular, etc.), rounded shape, circular shape, etc.

[0047] In example embodiments, each of the first contact arm portion **103A_c** and second contact arm portion **103B_c** may have one or more holes **103 H**. The one or more holes **103 H** may be configured to retain a fastener that holds a pad **106** against each of the first contact arm portion

103A_c and the second contact arm portion **103B_c**. The fastener may include bolts, rods, screws, etc. The pad **106** may be comprised of rubber, plastic, or the like, and may allow for the arms of an existing lift previously installed on the transportation vehicle to slide along the first contact arm portion **103A_c** and the second contact arm portion **103B_c**.

[0048] FIG. 5 is a view illustrating attachment and operation of lift safety mechanism, according to certain example embodiments. Detailed here is the relationship between the lift plate **102**, fixed plate **101**, and the first and second arms **103A** and **103B** with respect to the existing accessible lift mechanism **400** and the pressure plate **410**. As seen, the lift plate **102** and fixed plate **101** sit above the existing pressure plate **410** with the first and second arms **103A** and **103B** nested beneath respective lift arms **405** of the existing accessible lift mechanism **400**. The fixed plate **101** may be fastened to the vehicle using the original pressure plate fastener **415** inserted through the $\frac{3}{8}$ " hole in the fixed plate **101** with a rubber washer inserted beneath the existing pressure plate **410** and the steel channel below. As illustrated in FIG. 5, the pressure plate **410** may be mounted on the floor **415** of the transportation vehicle.

[0049] Each of the first lift arm **103A** and the second lift arm **103B** may nest beneath respective arms of an existing lift previously installed on the transportation vehicle. Each of the first contact arm portion **103A_c** and second contact arm portion **103B_c** may be configured to be in contact with the respective lift arms **405** of the existing lift **400**, such that when the lift arms **405** of the existing lift **400** extend to lower the floor of the existing lift **400**, the lift arms **405** of the existing lift **400** may press against the first contact arm portion **103A_c** and the second contact arm portion **103B_c** to cause the lift plate **102** to rise up away from the floor of the vehicle. For example, in operation, the lift arms **405** of the existing accessible lift mechanism **400** may extend forward and lower such that the lift arms **405** press against the first contact arm portion **103A_c** and the second contact arm portion **103B_c**, respectively. As the lift arms **405** press against the first contact arm portion **103A_c** and the second contact arm portion **103B_c**, respectively, the first and second arms **103A** and **103B** may cause the lift plate **102** to rise up. For example, an angle between the upper surface of the lift plate **102** and the upper surface of the fixed plate **101** may decrease as the first and second arms **103A** and **103B** may cause the lift plate **102** to rotate with respect to the hinge **105**. For example, the lift plate **102** may be raised to create an angle between the floor of the vehicle and the lift plate **102**, thereby preventing a wheelchair or the like from rolling beyond the lift plate **102**. For example, when in a raised position, the lift plate **102** may form a barrier to stop a wheelchair from rolling out of the transportation vehicle.

[0050] The invention is used in a transportation vehicle (e.g., vans, shuttles, buses) equipped with an accessible lift used to raise and lower wheelchairs/stretchers/individuals into and out of such vehicles. The mechanism is intended to prevent wheelchairs from rolling or being rolled out of the vehicle while the lift is deployed to its lowest position. As the lift is deployed, the safety plate raises to create a barrier preventing wheelchairs from rolling out of the vehicle.

[0051] While the disclosed embodiments describe the invention in connection with transportation vehicles, such as vans, shuttles, and buses, the disclosed embodiments may be configured for use in any other location in which a wheelchair lift may be employed.

Claims

1. A lift safety apparatus, comprising: a fixed plate configured to be removably attached within a vehicle; a lift plate adjacent to the fixed plate; and a hinge connecting the lift plate to the fixed plate, wherein the lift plate comprises first and second arms, and wherein each of the first and second arms comprises a first vertical arm portion, a second vertical arm portion, and a contact arm portion.
